Announcements – Jan 24

- HW 1 is due Friday at 9am.
- TA offices hours are posted. TAs OH in GS - 837
- SI sessions are Tue 6-7pm and Thu 2-3pm. Rm GS 228
- Quiz 4 trivia – The Osborne I weighed 25lbs. It was not exactly a laptop, but it was the first portable computer. It looked like a big plastic suitcase. 41% of you got this answer correct. 25lbs was the most common answer.

Quantification

Idea: Establish truth of predicates over sets of values.

Two common generalizations:

(1) Universal Quantification ($\forall$)

- **Reads "for every" or "for all"**

(2) Existential Quantification ($\exists$)

- **Reads "There exists"**

$\exists!$ is another quantification

- **Reads "There exists a unique"**

Another view

Suppose the domain of a predicate P(x) is $\{a_1, a_2, \ldots, a_n\}$

$\forall x\, P(x) \equiv P(a_1) \land P(a_2) \land \ldots \land P(a_n)$

$\exists x\, P(x) \equiv P(a_1) \lor P(a_2) \lor \ldots \lor P(a_n)$

Examples

Suppose P(x) is: x is a student named Luke

Is P(x) a proposition?

No, because whether it is true or false depends on a variable.

So is $\forall$ x P(x) a proposition?

Yes, but . . . the domain of P must be defined

Say the domain of P is all students in this class

Is $\forall$ x P(x) true or false?

Is $\exists$ x P(x) true or false?

Is $\exists!$ x P(x) true or false?

Examples

P(x): x has feathers and the domain is birds

Let's translate all the quantifiers to English

$\forall x P(x)$

Every bird has feathers

$\exists x P(x)$

There is a bird that has feathers

Quantified Statements

We can add quantifiers to compound propositions:

$S(x)$: x is a secret agent
$P(x)$: x is a platypus
Domain: all real life and fictional characters

$\exists x\ (S(x) \wedge P(x))$
There is a secret agent who is also a platypus

This is true.

$\exists x\ (S(x) \wedge \neg P(x))$
There is a secret agent who is not a platypus

This is also true.

Quantified Statements

A proposition that includes a quantifier is called a <u>quantified statement</u>

Note that quantifiers have higher priority than logical operators.

so $\exists\ x\ S(x) \land P(x)$
is evaluated as $(\exists\ x\ S(x)) \land P(x)$

This not a proposition.

The variable in S(x) is called a <u>bound variable</u>
because it's bound to a quantifier.

The variable in P(x) is called a <u>free variable</u>

Usually, we would use different labels for these variables to avoid confusion.
e.g. $\exists\ x\ S(x) \land P(y)$

Quantified Statements

The part of a logical expression to which a quantifier is applied is called the scope of that quantifier.

For example in:

$\forall x\big(P(x) \ \rightarrow Q(y)\big) \wedge R(x)$

the scope of $\forall x$ is $P(x) \ \rightarrow Q(y)$

Examples

C(x) x has a cape
F(x) x can fly
B(x) x is bullet proof
The domain is superheroes

Express

There is a bullet proof superhero who can fly and has a cape.

Every flying superhero has a cape.

Every superhero flies, is bullet proof, or has a cape.

A superhero can fly or be bullet proof, but not both.

C(x) x has a cape
F(x) x can fly
B(x) x is bullet proof
The domain is superheroes

Express

There is a bullet proof superhero who can fly and has a cape.

$$\exists x \, (C(x) \land F(x) \land B(x))$$

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Every flying superhero has a cape.

$$\forall x\ (F(x) \rightarrow C(x))$$

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Express

There is a bullet proof superhero who can fly and has a cape.

$$\exists x\, (C(x) \wedge F(x) \wedge B(x))$$

Every flying superhero has a cape.

$$\forall x\, (F(x) \rightarrow C(x))$$

Every superhero flies, is bullet proof, or has a cape.

$$\forall x\, (F(x) \vee B(x) \vee C(x))$$

A superhero can fly or be bullet proof, but not both.

$C(x)$ x has a cape
$F(x)$ x can fly
$B(x)$ x is bullet proof
The domain is superheroes

Express

There is a bullet proof superhero who can fly and has a cape.

$$\exists x\,(C(x) \land F(x) \land B(x))$$

Every flying superhero has a cape.

$$\forall x\,(F(x) \rightarrow C(x))$$

Every superhero flies, is bullet proof, or has a cape.

$$\forall x\,(F(x) \lor B(x) \lor C(x))$$

A superhero can fly or be bullet proof, but not both.

$$\forall x\,(\neg(F(x) \land B(x)))$$

So P(x): x is red, domain is types of Birds
is a function, it is not true or false as it is

P(blue jay) blue jay is red   Does evaluate to T or F

Adding quantifiers also makes the statement evaluate to T or F

$\exists x P(x)$ "There is a red bird" also evaluates to T or F

Examples

$P(x)$ is $x + 1 = 2$

is

$\exists x\, P(x)$ true ?

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We don't know! What is the domain?
If the domain is all integers, then it is true.
If the domain is all even integers, then it is false.

Examples

$P(x)$ is $x + 1 = 2$ Domain is all integers.

is

$\exists\, x\, P(x)$ true ?

Yes! $(x = 1)$

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Yes! $(x = 1)$

how about

$\forall\, x\, P(x)$?

False. $(x = 0)$

Example: Universal Quantification

Consider this conversational English statement:

All of Dr. Anson's students are geniuses.

How can we express that statement in logic notation?

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Easiest way: Let Domain be Dr. Anson's students

Define $\mathrm{P(x) = x\ is\ a\ genius}$

$\forall x\, P(x)$

Example: Universal Quantification

Consider this conversational English statement:

All of Dr. Anson's students are geniuses.

How can we express that statement in logic notation?

If we must use the domain All People

Define $P(x)$ = x is a genius
Define $Q(x)$ = x is a student of Dr. Anson

$\forall x\ (Q(x) \rightarrow P(x))$

Logical Equivalences Involving Quantifiers

Quantified statements are logically equivalent iff they have the same truth value no matter which predicates and domains are substituted into these statements.

e.g. $\forall x\big(P(x) \land Q(x)\big) \equiv \forall x P(x) \land \forall x Q(x)$

We will have better tools to prove this later. Think of this way. The first statement is true if for all x P(x) and Q(x) is true. This means for every x in the domain P(x) is true and Q(x) is true, so P(x) is true for every value and Q(x) is true for every value, so both $\forall x P(x) and\ \forall x Q(x)$ are true.

More DeMorgan

Suppose $P(x)$ is a predicate with domain $\{a_1, a_2, \ldots, a_n\}$

$\neg \forall x\, P(x) \equiv \neg(P(a_1) \wedge P(a_2) \wedge \ldots \wedge P(a_n))$

$\equiv \neg P(a_1) \vee \neg P(a_2) \vee \ldots \vee \neg P(a_n)$

$\equiv \exists x\, \neg P(x)$

DeMorgan also works if the domain is infinite, so:

$\neg \forall x\, P(x) \equiv \exists x\, \neg P(x)$

More DeMorgan

Suppose $P(x)$ is a predicate with domain $\{a_1, a_2, \ldots, a_n\}$

$$\neg \exists x\, P(x) \equiv \neg(P(a_1) \vee P(a_2) \vee \ldots \vee P(a_n))$$

$$\equiv \neg P(a_1) \wedge \neg P(a_2) \wedge \ldots \wedge \neg P(a_n)$$

$$\equiv \forall x\, \neg P(x)$$

DeMorgan also works if the domain is infinite, so:

$$\neg \exists x\, P(x) \equiv \forall x\, \neg P(x)$$

More DeMorgan – in English

Not every secret agent is human

$\neg \forall x\, P(x)$ P(x): x is human, domain all secret agents

There is a secret agent who is not a human.

$\exists x\, \neg P(x)$

There is no rollercoaster that is not fun.

$\neg \exists x\, \neg P(x)$ P(x): x is fun, domain all rollercoasters

Every rollercoaster is fun.

$\forall x\, P(x) \quad \equiv \forall x\, \neg\neg P(x)$

Predicates with Multiple Variables

So far, all the predicates we've looked at have a single variable, but there's no reason we can't have multiple variables.

Examples:

P(x,y): x and y are brothers

We need domains for both vars. Here the domains for x and y might be the same.

Q(x,y): x is covered with y

The domain for x is all animals.
The domain for y is {scales, fur, feathers}

Q(fox, fur) is true, Q(lizard, feathers) is false

Of course, we could also have 3, 4, or even 255 variables. (i.e. any number)